

Probability Theory

Lecture Notes

Instructor: Dr. Nishant Chandgotia

Written By-

Kartick Samanta

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Student Kartick Samanta

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1.1 Motivation and Some Problems: Monty Hall and Sleeping Beauty Problem

Question 1. Sleeping Beauty Problem

Princess, Villain, (Powerful Drug), Princess will sleep 2 years straight. Villain tosses a fair coin, If Heads appear Princess wakes up once briefly at the end of 1 year, If fails $\rightarrow$ at the end of 2 years. In no of cases, the princess will have any information about time of memory of having woken up. Suppose the princess wakes up, You ask what is the Probability of Heads what should the princess answer?

Solution:

- $\frac{1}{2}$ No information exchanged so probability should be $\frac{1}{2}$, If repeated independently a million times, half chances each.
- $\frac{2}{3}$ There are possibilities of this. Proportion of waking during Heads is $\frac{2}{3}$, and $\dots \frac{1}{3}$
- $\frac{3}{4}$

Villain tosses once, H $\rightarrow$ Villain tosses again, Two Possibilities H and T, so,

- HH $\rightarrow \frac{1}{4}$
- HT $\rightarrow \frac{1}{4}$
- T $\rightarrow \frac{1}{2}$

Important

All the solutions above are wrong. The Above problem has no solution, Because it is ill posed, and this ill posedness is with respect to current Framework.

Question 2. Monty Hall Problem

Monty Hall Problem: Todo: Add figure. There are three doors: one hides a car and the other two hide goats. After you choose one door say A, the host opens another door

revealing a goat.

1. Question: Should you switch?
2. Probability car behind C > Probability car behind A.

Solution: Yes, we should change our decision,

Initially, the probability that your chosen door contains the car is $P(\text{chosen door}) = \frac{1}{3}$, while the probability that the car is behind one of the other two doors is $P(\text{other two doors}) = \frac{2}{3}$.

Since the host always reveals a goat, this entire probability concentrates on the remaining unopened door. Therefore, $P(\text{Win by staying}) = \frac{1}{3}$, $P(\text{Win by switching}) = \frac{2}{3}$. Hence, switching gives twice the probability of winning compared to staying.

Summary

- Sleeping Beauty Problem
- Monty Hall Problem

End of Lecture

References

- [AD00] Robert B. Ash and Catherine A. Doléans-Dade. *Probability and Measure Theory*. 2nd ed. Academic Press, 2000.
- [Bil95] Patrick Billingsley. *Probability and Measure*. 3rd ed. John Wiley & Sons, 1995.
- [Cha26] Nishant Chandgotia. *Personal Website*. <https://nishantchandgotia.github.io/>. Accessed: 2026-08-05. 2026.
- [Dur19] Rick Durrett. *Probability: Theory and Examples*. 5th ed. Cambridge: Cambridge University Press, 2019.
- [Fel68] William Feller. *An Introduction to Probability Theory and Its Applications*. 3rd ed. Vol. 1. John Wiley & Sons, 1968.
- [GS12] Charles M. Grinstead and J. Laurie Snell. *Introduction to Probability*. 2nd ed. Available online. American Mathematical Society, 2012.
- [Gut13] Allan Gut. *Probability: A Graduate Course*. 2nd ed. Springer, 2013.
- [Hai03] John H. Haigh. *Taking Chances: Winning with Probability*. 2nd ed. Also published under the title “Knowing the Odds” in some editions/regions. Oxford University Press, 2003.
- [Ros19] Sheldon Ross. *A First Course in Probability*. 10th ed. Pearson, 2019.
- [Wil91] David Williams. *Probability with Martingales*. Cambridge University Press, 1991.